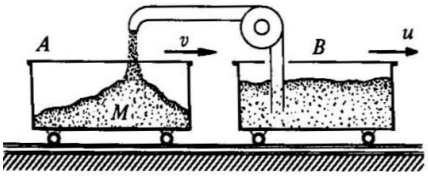
MIT 8.012 Problem Set 4 - Problem 3.11 (K&K)



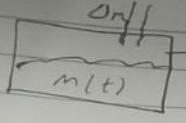
$t = 10$ s

3.11 Material is blown into cart A from cart B at a rate b kilograms per second. The material leaves the chute vertically downward, so that it has the same horizontal velocity as cart B, u. At the moment of interest, cart A has mass M and velocity v, as shown. Find dv/dt, the instantaneous acceleration of A.

Solution (handwritten) follows on the next pages.

Problem 2.62 (3rd Edition)

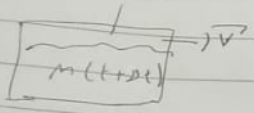
State 1 time t:



Notation: $\Delta m = M(t+\Delta t) - M(t)$

$\overrightarrow{p_{sys}}(t) = M(t)\,v(t) + \Delta m\, u$

State 2. time $t+\Delta t$



$\overrightarrow{p_{sys}}(t+\Delta t) = v(t+\Delta t)\, M(t+\Delta t)$

$= v(t+\Delta t)(M(t)+\Delta m)$

$\overrightarrow{F_{ext}} = 0$

$$\lim_{\Delta t \to 0} \frac{\overrightarrow{p_{sys}}(t+\Delta t) - \overrightarrow{p_{sys}}(t)}{\Delta t}$$

$$\lim_{\Delta t \to 0} \frac{v(t+\Delta t)\,M(t) + \Delta m\, v(t+\Delta t) - M(t)\,v(t) - \Delta m\, u}{\Delta t}$$

$$\lim_{\Delta t \to 0} M(t)\left(\frac{v(t+\Delta t) - v(t)}{\Delta t}\right) + \frac{\Delta m}{\Delta t}\left(v(t+\Delta t) - u\right)$$

$$M(t)\frac{dv}{dt} + \frac{dM}{dt}(v - u)$$

$$M(t)\frac{dv}{dt} + \frac{dM}{dt}(v-u) = 0$$

$$(bt+m_0)\frac{dv}{dt} = -b(v-u)$$

$$\frac{-1}{b}\int \frac{dv}{(v-u)} = \int \frac{dt}{(bt+m_0)}$$

$$\frac{-1}{b}\ln(|v-u|) = \frac{\ln(|bt+m_0|)}{b} + C$$

$$(|v-u|)^{-\frac{1}{b}} = c_0(bt+m_0)^{\frac{1}{b}}$$

$$\frac{1}{|v-u|} = c_0(bt+m_0)$$

$$= ut$$